

From Slope to Derivative

These five words look like five topics. They are **one story**, and they all answer the same question: **how steep is the graph here?** Read the chain below, then follow it one step at a time.

ЎЗБ Bu beshta soʻz bitta savolga javob beradi: grafik shu nuqtada qanchalik tik?

РУС Все пять слов отвечают на один вопрос: насколько круто идёт график в этой точке?

01 Slope Steepness of a straight line. burchak koeffitsiyenti · наклон	02 Gradient The same number — but a curve needs it <i>at a point</i> . burchak koeffitsiyenti · угловой коэффициент	03 Secant line Cuts the curve twice. Gives average steepness. kesuvchi · секущая	04 Tangent line Touches at one point. Gives steepness there. urinma · касательная	05 Derivative The number that is the gradient of that tangent. hosila · производная
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01 Slope of a straight line

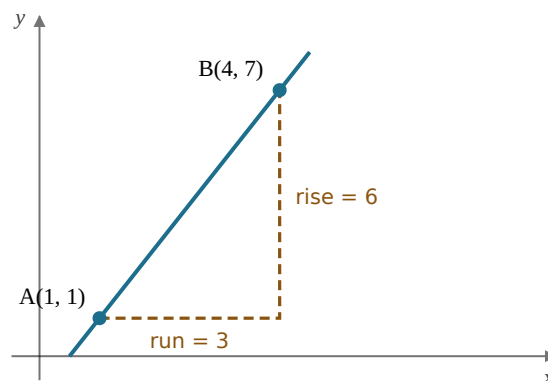
toʻgʻri chiziqning burchak koeffitsiyenti · наклон прямой

Take any two points on a straight line. Go **across**, then go **up**. Divide.

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6}{3} = 2$$

Slope 2 means: every 1 step to the right, the line climbs 2 steps up.

A straight line is equally steep everywhere. Pick *any* two points on it and you get the same answer — this is exactly what makes a line straight.



Slope of the line through A and B: rise 6 divided by run 3.

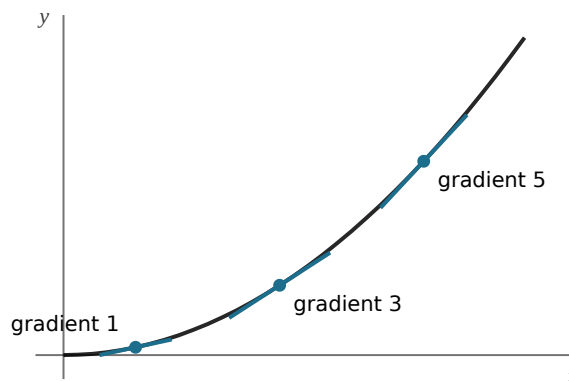
02 Gradient — same number, harder question

egri chiziqning qiyaligi · крутизна кривой

Gradient is another word for slope. In your textbook it is the *burchak koeffitsiyenti* / угловой коэффициент.

Now look at a curve. It is gentle near the bottom and steep further along. So this question has no answer: “What is the gradient of $y = x^2$?”

Only this one does: **“What is the gradient of the curve at this point?”** Everything that follows exists to answer it.



One curve, three points, three different gradients: 1, 3 and 5.

03 Secant line — steepness between two points

kesuvchi · секущая

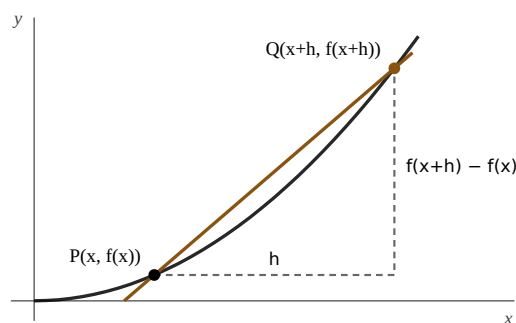
We only know how to find the slope of a *straight* line. So draw one: take two points on the curve, P and Q, and join them. That line is the **secant**.

Call the step across h . Then P is $(x, f(x))$ and Q is $(x+h, f(x+h))$, and the slope is:

$$\frac{f(x+h) - f(x)}{h}$$

For $f(x) = x^2$ with P(1, 1) and Q(3, 9): $(9 - 1) / (3 - 1) = 4$.

What does 4 mean? It is the **average** steepness from $x = 1$ to $x = 3$ — like average speed on a journey: total distance over total time. It says nothing about one single moment.



The secant through P and Q. Its slope is the step up divided by the step across, h .

04 Tangent line — steepness at one point

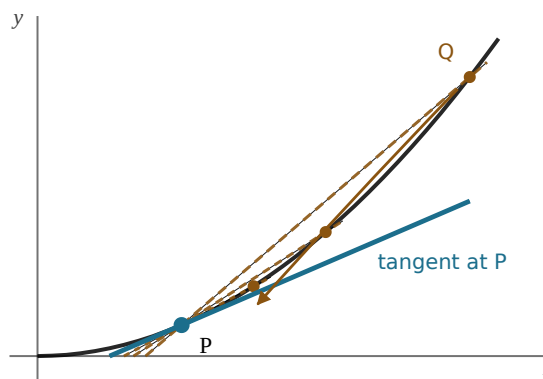
urinma · касательная

Now we want the steepness *exactly* at P. One point is not enough: the rise is 0 and the run is 0, and 0/0 is meaningless.

So keep two points — but slide Q down the curve towards P, and watch the secant slopes. For $f(x) = x^2$ at $x = 1$:

H	1	0.5	0.1	0.01	0.001	$\rightarrow 0$
SECANT SLOPE	3	2.5	2.1	2.01	2.001	2

The numbers do not jump around. They walk steadily towards **2**. And the secant lines themselves settle onto one line: the **tangent** at P. It touches the curve at P and points in the same direction the curve is going there.



Three secants (brown, dashed) as Q slides towards P. They close on the tangent (blue).

CAREFUL “Touches at one point” describes the direction near P. A tangent is still allowed to cross the curve somewhere far away — that does not stop it being a tangent at P.

05 Derivative — the limit gets a name

hosila · производная

“The number the secant slopes walk towards” is a mouthful. Mathematics gives it a name and a symbol — the **derivative**:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Do it once by hand for $f(x) = x^2$, so the table above stops being magic:

$$\frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h$$

We may cancel h because $h \neq 0$ — Q has not reached P yet, it is only close.

Now let $h \rightarrow 0$: $f'(x) = 2x$, so $f'(1) = 2$.

The same 2 the table was walking towards — found exactly, not guessed.

THE WHOLE CHAIN IN ONE SENTENCE

The derivative is the gradient of the tangent — and the tangent is what the secant becomes when the two points meet.

ЎЗБ Hosila — urinmaning burchak koeffitsiyenti; urinma esa ikki nuqta birlashganda kesuvchidan hosil bo'ladi.

РУС Производная — угловой коэффициент касательной; а касательная — это то, во что превращается секущая, когда две точки сливаются.

Words you will hear in three languages

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
straight line	to'g'ri chiziq	прямая
curve	egri chiziq	кривая
slope / gradient	burchak koeffitsiyenti, qiyalik	угловой коэффициент, наклон
rise (step up), Δy	funksiya orttirmasi, Δy	приращение функции, Δy
run (step across), Δx	argument orttirmasi, Δx	приращение аргумента, Δx
secant line	kesuvchi	секущая
tangent line	urinma	касательная
point of tangency	urinish nuqtasi	точка касания
difference quotient	ayirmali nisbat	разностное отношение
limit	limit	предел
derivative	hosila	производная
average rate of change	o'rtacha o'zgarish tezligi	средняя скорость изменения
instantaneous rate of change	oniy o'zgarish tezligi	мгновенная скорость изменения

Check yourself · O'zingizni tekshiring · Проверьте себя

1. A straight line passes through (2, 3) and (6, 11). Find its slope. Then find it again using two *different* points on the same line — what do you notice?
2. For $f(x) = x^2$, find the slope of the secant between $x = 2$ and $x = 2.1$.
3. Using $f(x) = 2x$, write down the gradient of the tangent at $x = 2$. Compare it with your answer to question 2. Why are they close but not equal?

ANSWERS 1 — slope 2, and any two points give 2, because the line is equally steep everywhere. 2 — $(4.41 - 4) / 0.1 = 4.1$. 3 — the tangent gradient is 4. The secant used a step of $h = 0.1$, so it gives the average over that step, slightly steeper than the value at the point itself. Shrink h and 4.1 walks towards 4.